

Cumulative H01 + A02 +
A05 + A06 source
same code and runtime
command

PGO is reproducible by
other builders
regenerate for exact
source and compiler
use a representative
pinned training recipe

PGO evaluation

Instrumented build
full public-corpus training
1:20:21 wall time
163 profile files

Profile-use build
mismatch is an error
binary: 3.57 MB to 2.78 MB
22.15% smaller

Five-pair median
55.89 s to 55.20 s
1.20% gain
5 of 5 outputs passed

REJECTED
below 2% practical gate

LTO evaluation

GCC 13.3 LTO build
cross-file optimization
binary: 3.57 MB to 2.95 MB
17.47% smaller

Five-pair median
55.75 s to 55.32 s
1.21% gain
5 of 5 outputs passed

REJECTED
below 2% practical gate

No toolchain variant
stacked
cumulative source remains
H01 + A02 + A05 + A06